Crystal Lattices as Markov Blankets

Introduction

This module explores **Systems** within the context of **Metallurgical Systems**. In the Active Inference framework, systems plays a critical role in how systems maintain their identity, process information, and adapt to perturbation. Here we examine this through the lens of crystal structures, defects, and fundamental thermodynamics.

Key themes: Unit cells, crystal boundaries, Bravais lattices, atomic-scale system boundaries

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Learning Objectives

By the end of this module, you will be able to:

1. Define **Systems** within the Active Inference framework as it applies to metallurgical systems
2. Identify how systems manifests in crystal structures, defects, and fundamental thermodynamics
3. Connect the formal FEP concept of systems to specific metallurgical phenomena
4. Analyze real-world case studies involving systems in materials engineering
5. Apply systems principles to solve practical metallurgical problems

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Key Concepts

1. Systems in the Active Inference Framework

In Active Inference, systems refers to the process by which systems maintain and update their relationship with the environment. For metallurgical systems, this maps directly onto physical processes: Unit cells, crystal boundaries, Bravais lattices, atomic-scale system boundaries.

The Free Energy Principle provides a unifying lens: every metallurgical phenomenon involving systems can be understood as a system minimizing the difference between its current state and its preferred (equilibrium) state.

2. Systems at the Atomic Scale

At the most fundamental level, systems in metallurgy operates through atomic-scale mechanisms. Atoms in a crystal lattice interact with their neighbors according to interatomic potentials, and the collective behavior of these interactions gives rise to macroscopic material properties.

3. Systems at the Mesoscale

Moving up in scale, systems manifests through microstructural features — grain boundaries, precipitates, phase interfaces, and defect networks. These mesoscale structures mediate between atomic-level events and engineering-scale properties.

4. Systems at the Engineering Scale

At the largest scale relevant to this module, systems is expressed through macroscopic material behavior and process-level phenomena. This is where the metallurgist's interventions — heat treatments, deformation processes, and compositional design — interact with the material's inherent tendencies.

5. The FEP Bridge: From Thermodynamics to Inference

The deepest insight of this curriculum is that thermodynamic free energy minimization and variational free energy minimization share the same mathematical structure. In the context of systems, this means that the physical processes governing metallurgical behavior are formally analogous to the inference processes governing adaptive systems.

Applications

Case Study: Systems in Steel Processing

Consider a plain carbon steel (Fe-0.4wt%C) undergoing heat treatment. The concept of systems manifests concretely: the material system processes information (temperature, composition gradients), updates its internal state (crystal structure, phase fractions), and acts to minimize its free energy through phase transformations.

Case Study: Systems in Aluminum Alloy Design

In Al-Cu precipitation-hardening alloys (such as 2024-T3), systems operates through the sequence of metastable precipitate formation: GP zones -> theta'' -> theta' -> theta (Al2Cu). Each stage represents the system's ongoing inference about its equilibrium state.

Cross-References

- For parallel treatment in other courses, see the Cross-Course Map
- See the Glossary for term definitions
- See the Notation Table for symbol conventions

Summary

Concept	Metallurgical Meaning
Systems	Unit cells, crystal boundaries, Bravais lattices, atomic-scale system boundaries
FEP Connection	Free energy minimization drives systems in both thermodynamic and inferential frameworks
Scale	Operates from atomic (angstrom) through mesoscale (micrometer) to engineering (meter)
Key Techniques	Characterization and modeling tools specific to systems in this context

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References

- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127-138.
- Porter, D. A., Easterling, K. E., & Sherif, M. Y. (2021). *Phase Transformations in Metals and Alloys* (4th ed.). CRC Press.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.
- Callister, W. D., & Rethwisch, D. G. (2020). *Materials Science and Engineering* (10th ed.). Wiley.